

PAPER_1682 — DM Direct-Detection Floor = $\sigma_{\text{floor}} = \Lambda^4 \times 10^{-40} \text{ cm}^2$ (predicts null detections)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Dark Matter **Date:** June 18, 2026 **Status:** CLOSED — EXACT formula closure (PAPER_145x-147x broader corpus)

Observation

PAPER_1454 (PAPER_14xx broader corpus, classic cosmology/BSM puzzle) gives:

$\sigma_{\text{floor}} = \Lambda^4 \times 10^{-40} \text{ cm}^2$ (predicts null detections)

Status: **EXACT formula.**

UQFF Closed Identity

$\sigma_{\text{floor}} = \Lambda^4 \times 10^{-40} \text{ cm}^2$ (predicts null detections) EXACT formula

The expression uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Empirical dark matter measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1454
- Related: PAPER_1666-1675 (tier-25); PAPER_1656-1665 (tier-24)
- Calculator dispatch: `calculate_paradox({"paradox": "dm_direct_floor_lambda4"})`

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